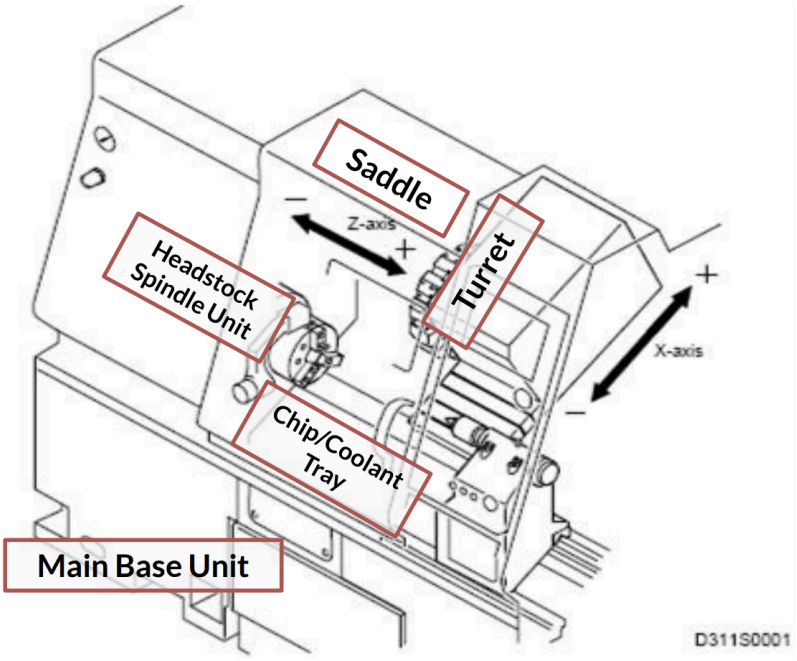
CNC Turning Fundamentals

CNC turning machines have 5 main elements:



Main Base Unit:

supports machine

holds motor, hydraulic tank, coolant pump and power transformer

Headstock Spindle Unit:

houses the spindle and chuck to hold and rotate the stock in either direction. mainly hard jaws are used, although soft jaws may be used too.

Saddle:

Moves turret and tool along the z axis, or longitudinally

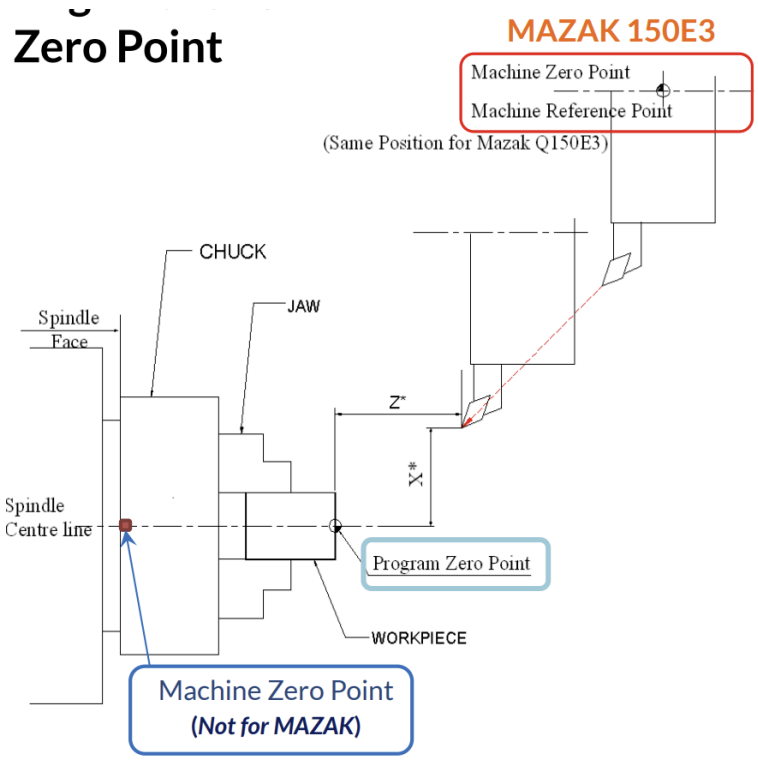
Turret

moves tool along x axis, radially. holds 8 tools. cutters at odd positions (1,3,5,7) are for internal diameter (ID) operations. cutters at even positions (2,4,6,8) are for outer diameter (OD) operations.

Chip & Coolant Tray

collects chips and coolant, movable

The MAZAK 150E3 machine has its zero point above the workpiece, unlike others which may be located at the spindle. This determines the default, or home, position of the turret.



For programming, the zero point is the rightmost end of the stock, aligned to the spindle. This is where the tool will first move to before it starts to take cuts from the stock.

To program CNC machines, we use NC (numerical controlled) programs.

Each program consists of blocks, which refer to each line. [N40 T0303]

each block contains words [N40], which is a string of characters separated by spaces.

Each word starts with a letter, called the address [N], and a number [40].

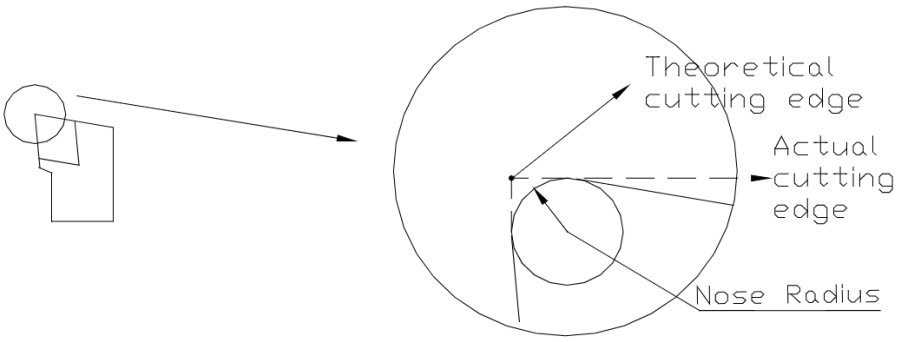
the NC syntax is as follows:

```
O1234 (program number, like the name of the program. commen
ts are brackets)
N10 (the N address acts as the line number)
N20 G0 G40 G21 (rapid positioning, nose radius compensation
cancel, metric units)
N30 G54 (workpiece coordinate system, N20 and N30 are defau
```

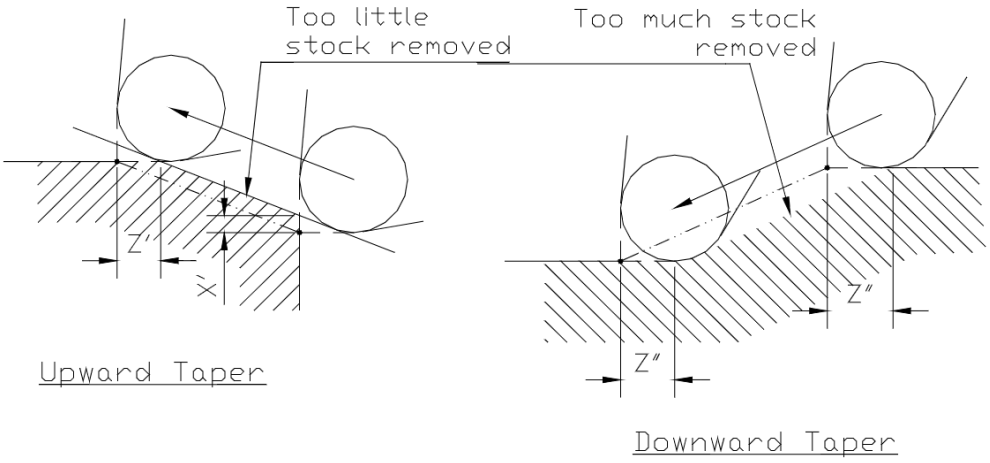
```
lt config settings)
N40 G28 U0. W0. (return to machine home position)
N50 T0303 (set to tool 3 with tool offset 3)
N60 G50 S2500 (set max spindle speed to 2500 rpm)
N70 G96 S90 M3 (set constant surface speed to 90m/min clock
wise[M3])
N80 G0 X12 Z2 M8 (move tool with rapid positioning to these
coords, with coolant on [M8])
N90 G1 X19 Z-1.5 F0.1 (linearly interpolate to this coord a
t feedrate 0.1 mm/rev)
. (turret will continue to linearly interpolate to the rest
of the coords)
.
.
N150 G0 X80 Z75 M9 (rapid position back to these coords, tu
rn coolant off)
N160 G28 U0. W0. M5 (return to machine home, turn spindle o
ff)
N170 G02 (end program)
```

Each G code syntax does not need to be memorised.

Recall that turning cutting tools have a nose radius, for better surface finish and prolonging tool life. However, CNC programming is based on the theoretical cutting edge/theoretical tip of the tool, as shown



Hence, depending on the path of the tool, the cutting tool may machine off too much or too little of the stock, causing dimensional inaccuracy.



To compensate for this, we use the tool nose radius compensation, programmed as G41 or G42.

G41 is used when the tool is on the left of the workpiece, viewed from the cutters movement.

G42 is used when the tool is on the right of the workpiece, viewed from the cutters movement.